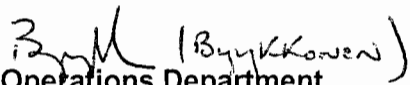


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Main Turbine		
TASK:	TCAF Main Turbine Trip < P-9 (Main Turb ASO trip)		
TASK NUMBER:	N1140420401		
JPM NUMBER:	11-01 NRC Sim e		
ALTERNATE PATH:	☒	K/A NUMBER:	2.4.31
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	S2.OP-AR.ZZ-0007, Rev. 48, Overhead Window G S2.OP-AB.TRB-0001, Rev. 14, Turbine Trip <P-9 (Both rev checked 8-6-12)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	5 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-6-12
Validated By:	Mulford SME or Instructor	Date:	8-23-12
Approved By:	 Training Department	Date:	9/12/12
Approved By:	 Operations Department	Date:	9/5/12
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	SAT	UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Main Turbine

TASK: TCAF Main Turbine Trip < P-9 (Loss of Turbine Auxiliaries Cooling Expansion Tank Level)

TASK NUMBER: N1140420401

SIMULATOR SETUP IC-255

MALF TU0118A 21MS28 Turb Stop Valve Fails Open

RT-2

I/O 4505 OVLO Low Oil Pressure – I

MALF AN0244 SER 244Fails- :G4 Turbine Auto Stop Oil Pressure Low

RT-3

MALF AN0605 SER 605 Fails - : F32 DEHC Trip

I/O 4506 OVLO Low Oil Pressure – II

I/O 4507 OVLO Low Oil Pressure – III

INITIAL CONDITIONS:

40% power, BOL.

A power reduction to bring the Main Turbine off-line is on hold.

INITIATING CUE: You are the Reactor Operator. Respond to all alarms and indications.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Trip the Main Turbine.
2. Trip the Rx.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Main Turbine

TASK: TCAF Main Turbine Trip < P-9 (Low ASO pressure)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		<u>SIMULATOR OPERATOR:</u> Insert RT-2 after candidate assumes the watch.	Announces unexpected OHA G-4 TURB AUTO STOP OIL PRESS LO		
			Refers to S2.OP-AR.ZZ-0007, Overhead Window G.		
	ARP 1.0/2.0		Reviews causes and determines <u>one</u> of <u>three</u> Auto Stop Oil Pressure channels reading ≤ 50 psig causes the alarm, and 2 of 3 channels in alarm will trip the Main Turbine.		
	ARP 3.1	IF alarm is due to testing OR alarm is expected for current plant conditions, THEN no further action is required.	Determines no testing in progress and alarm is not expected.		
	ARP 3.2	CONFIRM alarm by verifying 2RP4 AST Oil Press Low status light indications.	Checks 2RP4 AST Oil Press Low status light is illuminated for Channel I.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Main Turbine

TASK: TCAF Main Turbine Trip < P-9 (Low ASO pressure)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP 3.3	<u>IF</u> a Turbine Trip occurs, <u>THEN</u> : A. <u>IF</u> $\geq$ P-9 (49% power), THEN GO TO 2-EOP-TRIP-1, Reactor Trip or Safety Injection. B. <u>IF</u> <P-9 (49% power), THEN GO TO S2.OP-AB.TRB-0001(Q), Turbine Trip Below P-9.			
	ARP 3.4	IF instrument failure is suspected, THEN: A. REFER TO Technical Specifications. B. REMOVE affected channel from service IAW S2.OP-SO.RPS-0006(Q), Main Turbine Channel Trip/Restoration. C. INITIATE a NOTF to determine and correct the cause of the alarm.	There is no justification for suspecting instrument failure without having technician investigate. May dispatch operator to check turbine or call for assistance from WCC or maintenance. Note: Allow sufficient time for candidate to try and initiate action to investigate.		
			Simulator Operator: Insert RT-3 on direction from Evaluator. This simulates the remaining 2 channels see a low Auto Stop Oil signal and actuates OHA F-32, which indicates a demand for a Turbine Trip, which does not occur.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Main Turbine

TASK: TCAF Main Turbine Trip < P-9 (Low ASO pressure)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*			Candidate recognizes a demand for a turbine trip has occurred, but the turbine has not tripped. Initiates a turbine trip from trip handle and verifies turbine is tripped. Recognizes step 3.3 states... <u>"IF <P-9 (49% power),</u> <u>THEN GO TO S2.OP-AB.TRB-0001(Q),</u> Turbine Trip Below P-9." And goes to S2.OP-AB.TRB-0001.		
	AB.TRB 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Main Turbine

TASK: TCAF Main Turbine Trip < P-9 (Low ASO pressure)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	AB.TRB 3.2	VERIFY the Turbine is tripped.	Note: <u>IF</u> the candidate observes the 21MS28 remaining open at this point, THEN they may determine the turbine is not tripped, and perform a MSLI by depressing all 4 MSLI PBs on either Safeguards bezel. Initiating a MSLI before the Rx is tripped is an <u>incorrect action</u> which may challenge SG safeties if they lift. Initiating a MSLI renders the Main Steam Dumps inoperable and prematurely isolates the SGFPs. With Rx power >P-10 (10%) and steam dumps not available, a Rx trip is required IAW CAS actions 1.0 and 2.0. See page 8 for complete discussion. See <u>next page</u> for required actions in EOP-TRIP-1 IF a Rx trip is performed here.		
	AB.TRB 3.3	Are all Turbine Stop Valves closed?	Determines all Turbine Stop Valves are NOT closed from 2RP4. Goes to Step 3.7.		
	AB.TRB 3.7	TRIP the Reactor, GO TO 2-EOP-TRIP-1, Reactor Trip or Safety Injection.	Trips the Reactor using either of the Reactor Trip handles.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Main Turbine

TASK: TCAF Main Turbine Trip < P-9 (Low ASO pressure)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	EOP- TRIP-1 1	TRIP REACTOR	IF candidate initiates a Rx trip prior to initiating a MSLI based on CAS actions 1.0 and 2.0 which would occur if they initiated a MSLI, then successful performance of TRIP-1 immediate actions must be performed as follows: Trips the Rx using either Trip handle.		
	EOP- TRIP-1 2	IS REACTOR TRIP CONFIRMED	Confirms Rx trip		
*	EOP- TRIP-1 3	TRIP TURBINE	Attempts to trip the Main Turbine using the Trip handle. Determines the Trip handle did not close all Main Turbine Stop Valves. Depresses Turbine Trip bezel. Determines the Trip handle did not close all Main Turbine Stop Valves.		
*	EOP- TRIP-1 3	TRIP TURBINE (MSLI step)	Initiates a MSLI. Terminate JPM after MSLI has been performed.		

Turbine Trip requirements- During performance of this JPM, the candidate will receive direction from the Alarm Response Procedure S2.OP-AR.ZZ-0007, Overhead Window G, OHA G-4, page 9, which states “**TRIP** the Turbine and GO TO S2.OP-AB.TRB-0001, Turbine Trip <P-9.”

When outside the EOP network, initiating a MSLI to perform a Turbine Trip is NOT an appropriate action, since in TRIP-1 the steps for tripping the Rx precede the steps for tripping the Turbine, and the MSLI would only be performed AFTER all attempts to trip the Rx from the control room have already been performed. IF the Rx trip were NOT confirmed, then the Turbine trip steps would include the MSLI since the overriding concern at that point would be reduction of Rx power by removing the Main Turbine from service.

After entering the EOP Network, actions for tripping the Reactor and tripping the Turbine have special definitions as described in OP-AA-101-111-1003, Use of Procedures, which **ONLY** apply while in the EOP network. When defining turbine trip while in EOP-TRIP-1, it states on page 13,

“ Steps 2.2 and 3, **TRIP TURBINE**, are defined as,

“Operate the turbine trip switch”

IF turbine trip NOT confirmed,

THEN trip the turbine using the Turbine Trip bezel.

IF turbine trip NOT confirmed,

THEN initiate main steam isolation.””

For these reasons, initiating a MSLI during this JPM **prior** to initiating a Rx trip will be considered failure criteria due to the possible challenge to SG safeties which may occur due to this action, and which otherwise would not occur in the JPM.

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- MM 1. Task description and number, JPM description and number are identified.
- MM 2. Knowledge and Abilities (K/A) references are included.
- MM 3. Performance location specified. (in-plant, control room, or simulator)
- MM 4. Initial setup conditions are identified.
- MM 5. Initiating and terminating Cues are properly identified.
- MM 6. Task standards identified and verified by SME review.
- MM 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- MM 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 48/14 Date 8/23/12
- MM 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- N/A 10. ☒ If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

SME/Instructor: _____

SME/Instructor: _____

Date: _____

Date: _____

Date: _____

INITIAL CONDITIONS:

40% power, BOL.

A power reduction to bring the Main Turbine off-line is on hold.

INITIATING CUE:

You are the Reactor Operator. Respond to all alarms and indications.